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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
accuracy = accuracy_score(y_test, y_pred)
print("Naive Bayes Classification")
print("-----")
print("Training Samples :", len(X_train))
print("Testing Samples  :", len(X_test))

print("\nConfusion Matrix:")
print(cm)

print("\nAccuracy :", round(accuracy * 100, 2), "%")

```

... Naive Bayes Classification

Training Samples : 105

Testing Samples : 45

Confusion Matrix:

```

[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

```

Accuracy : 97.78 %
